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##COIT20246 Cyber Security and Networking##

Project Specification 1

Small Business Network Security, OpenWrt Firewall Configuration and Cyber Security Risk Assessment

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Business Scenario: Small IT and Business Consultancy, Brisbane, Queensland, Australia

4.1 Network Setup

4.1.1 List the Assumptions

Location: The business is assumed to operate in Brisbane, Queensland, Australia.

Professional services: It provides IT support, cybersecurity consulting, network configuration and business technology services.

Staff: 8 staff -- 1 Business Manager, 2 IT Support Specialists, 3 Business Consultants and 2 Administrative Staff.

Business/public information: The Website contains business and service information, contact information, project information, and Identification information of staff members/students needed for testing.

4.1.2 Set Up the Network

```

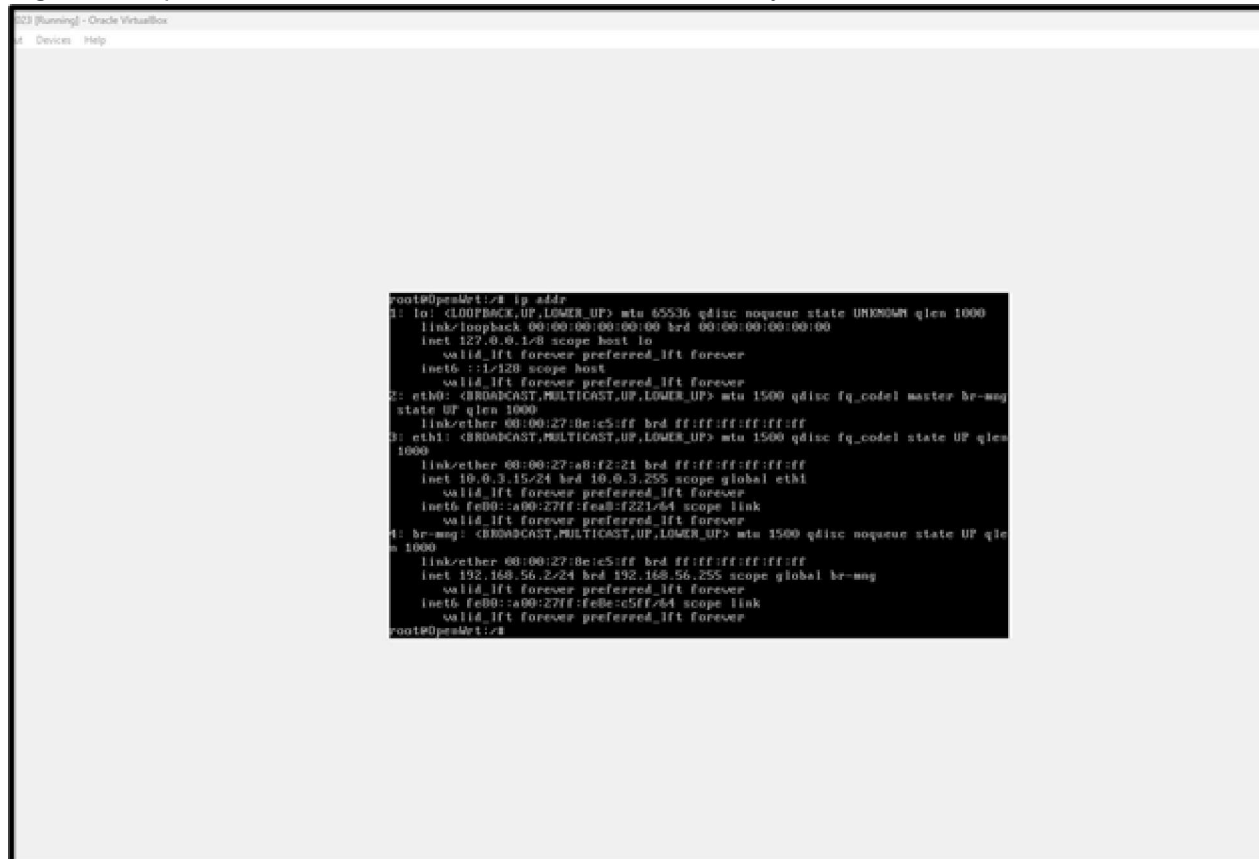
9.181430] br-mng: port 1(eth0) entered disabled state
9.188530] device eth0 entered promiscuous mode
9.216545] 8021q: adding VLAN 0 to HW filter on device eth1
11.221501] e1000: eth0 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: R
X
11.227515] br-mng: port 1(eth0) entered blocking state
11.230907] br-mng: port 1(eth0) entered forwarding state
11.234240] IPv6: ADDRCONF(NETDEV_CHANGE): br-mng: link becomes ready
11.302330] e1000: eth1 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: R
X
11.308511] IPv6: ADDRCONF(NETDEV_CHANGE): eth1: link becomes ready
S

BusyBox v1.35.0 (2023-01-03 00:24:21 UTC) built-in shell (ash)

_ _ _ _ _ 
| |   | |   | |   | |   | |   | |   | |   | |   | |   | |   | |   | |
|_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_|
      W I R E L E S S F R E E D O M

-----
OpenWrt 22.03.3, r20020-43d7ad93e
-----
root@OpenWrt:~# S_
```

Figure 1: OpenWrt 22.03.3 Virtual Machine Boot and System Information



```

root@OpenWrt:~# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master br-mag state UP qlen 1000
    link/ether 00:00:27:0e:c5:ff brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP qlen 1000
    link/ether 00:00:27:a0:f2:21 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe00:f221/64 scope link
        valid_lft forever preferred_lft forever
4: br-mag: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP qlen 1000
    link/ether 00:00:27:0e:c5:ff brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.2/24 brd 192.168.56.255 scope global br-mag
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe00:c5ff/64 scope link
        valid_lft forever preferred_lft forever
root@OpenWrt:~#

```

Figure 2: OpenWrt Network Interface Configuration and IP Address Allocation



```

ip addr add|del IFADDR dev IFACE ; show|flush [dev IFACE] [to PREFIX]
ip route list|flush|add|del|change|append|replace|test ROUTE
ip link set IFACE [up|down] [arp on|off] [multicast on|off]
    [promisc on|off] [mtu NUM] [name NAME] [qlen NUM] [address MAC]
    [master IFACE ; nomaster] [netns PID]
ip neigh show|flush [to PREFIX] [dev DEV] [nud STATE]
ip rule [list] ; add|del SELECTOR ACTION
root@OpenWrt:~# ubus call system board
{
  "kernel": "5.10.161",
  "hostname": "OpenWrt",
  "system": "11th Gen Intel(R) Core(TM) i5-11400F @ 2.60GHz",
  "model": "innotek GmbH VirtualBox",
  "board_name": "innotek-gmbh-virtualbox",
  "rootfs_type": "ext4",
  "release": {
    "distribution": "OpenWrt",
    "version": "22.03.3",
    "revision": "r20028-43d71ad93e",
    "target": "x86_64",
    "description": "OpenWrt 22.03.3 r20028-43d71ad93e"
  }
}
root@OpenWrt:~#

```

Figure 3: OpenWrt System and Virtual Machine Information

```

valid_lft forever preferred_lft forever
inet6 fe80::a00:27ff:fe80:221:64 scope link
valid_lft forever preferred_lft forever
br-mng: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP qlen 1000
    link/ether 00:00:27:0e:c5:ff brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.2/24 brd 192.168.56.255 scope global br-mng
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe80:c5ff:64 scope link
        valid_lft forever preferred_lft forever
root@OpenWrt:~# ip -br addr
BusyBox v1.35.0 (2023-01-03 00:24:21 UTC) multi-call binary.

Usage: ip [OPTIONS] address|route|link|neigh|rule [ARGS]

OPTIONS := -f{amily} inet|inet6|link | -o{uteline}

ip addr add|del IFADDR dev IFACE | show|flush {dev IFACE| {to PREFIX}
ip route list|flush|add|del|change|append|replace|test ROUTE
ip link set IFACE {up|down} {arp on|off} {multicast on|off}
    {promisc on|off} {mtu NUM} {name NAME} {qlen NUM} {address MAC}
    {master IFACE | nomaster} {netns PID}
ip neigh show|flush {to PREFIX} {dev DEV} {nud STATE}
ip rule {list} | add|del SELECTOR ACTION
root@OpenWrt:~#

```

Figure 4: OpenWrt Network Configuration and Interface Mapping

```

network.loopback.ipaddr='127.0.0.1'
network.loopback.netmask='255.0.0.0'
network.globals=globals
network.globals.ula_prefix='fd88:3563:c944::/48'
network.@device[0]=device
network.@device[0].name='br-lan'
network.@device[0].type='bridge'
network.@device[0].ports='eth0'
network.wan=interface
network.wan.device='eth1'
network.wan.proto='dhcp'
network.wan.ifname='eth1'
network.mng=interface
network.mng.proto='static'
network.mng.netmask='255.255.255.0'
network.mng.ipaddr='192.168.56.2'
network.mng.device='br-mng'
network.lan=interface
network.lan.proto='dhcp'
network.lan.device='eth2'
network.@device[1]=device
network.@device[1].name='br-mng'
network.@device[1].type='bridge'
network.@device[1].ports='eth0'
root@OpenWrt:~#

```

Figure 5: OpenWrt Network Interface and DHCP Configuration

```
network.mng.device='br-mng'
network.lan=interface
network.lan.proto='dhcp'
network.lan.device='eth2'
network.@device[1]=device
network.@device[1].name='br-mng'
network.@device[1].type='bridge'
network.@device[1].ports='eth0'
root@OpenWrt:~# ip route
default via 10.0.3.2 dev eth1 src 10.0.3.15
10.0.3.0/24 dev eth1 scope link src 10.0.3.15
192.168.56.0/24 dev br-mng scope link src 192.168.56.2
root@OpenWrt:~# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master br-mng
    state UP qlen 1000
    link/ether 08:00:27:8e:c5:ff brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP qlen
    1000
    link/ether 08:00:27:a8:f2:21 brd ff:ff:ff:ff:ff:ff
4: br-mng: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP qlen
    1000
    link/ether 08:00:27:8e:c5:ff brd ff:ff:ff:ff:ff:ff
root@OpenWrt:~# _
```

Figure 6: OpenWrt Routing Table and Network Interface Status

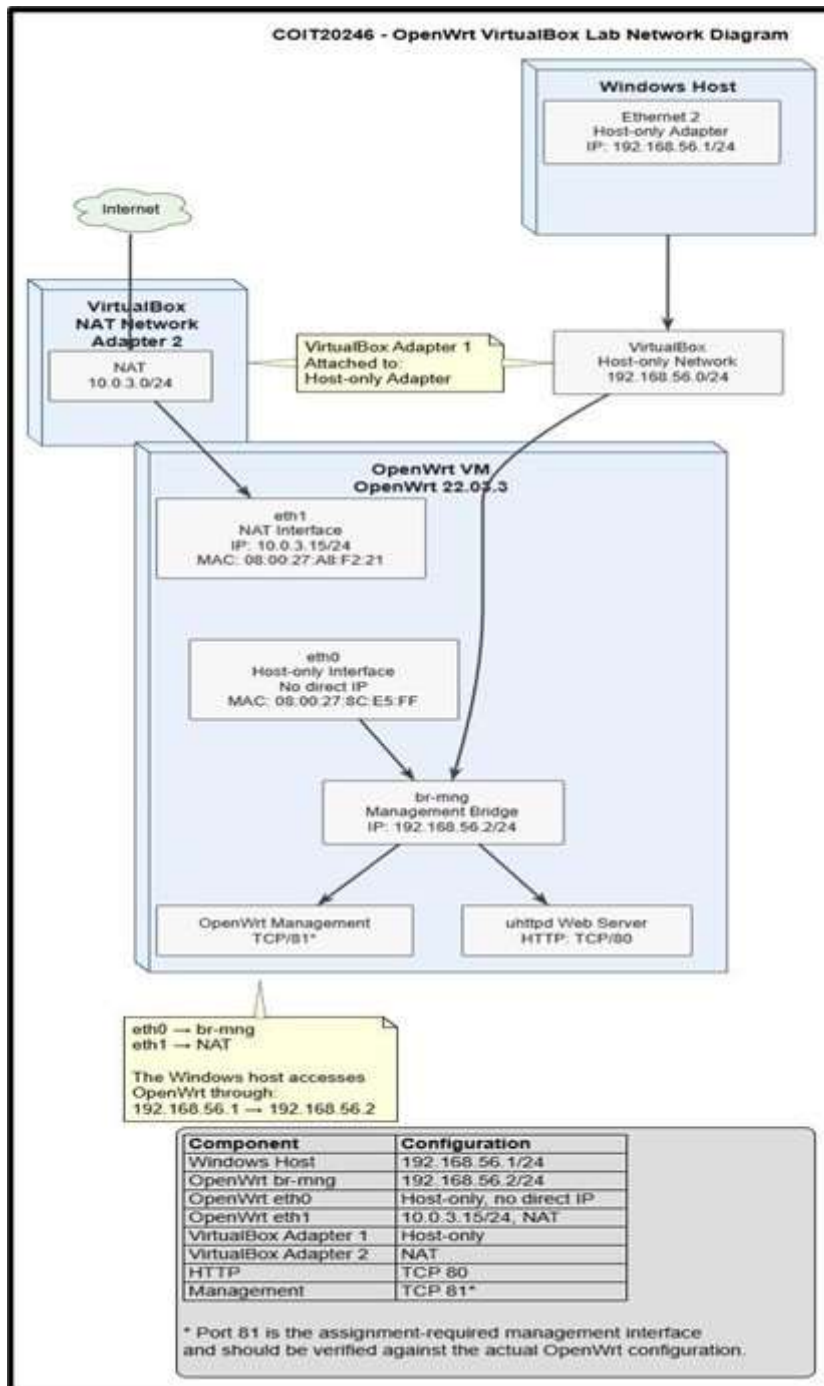


Figure 7: Proposed OpenWrt VirtualBox Laboratory Network Topology and IP Address

```
C:\Users\admin>ping 192.168.56.2

Pinging 192.168.56.2 with 32 bytes of data:
Reply from 192.168.56.2: bytes=32 time<1ms TTL=64
Reply from 192.168.56.2: bytes=32 time<1ms TTL=64
Reply from 192.168.56.2: bytes=32 time<1ms TTL=64
Reply from 192.168.56.2: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.56.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\admin>
```

Allocation

Figure 8: Successful ICMP Connectivity Test from Windows Host to OpenWrt

```

root@OpenWrt:~#
root@OpenWrt:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: seq=0 ttl=255 time=27.300 ms
64 bytes from 8.8.8.8: seq=1 ttl=255 time=27.892 ms
64 bytes from 8.8.8.8: seq=2 ttl=255 time=26.690 ms
64 bytes from 8.8.8.8: seq=3 ttl=255 time=28.318 ms
64 bytes from 8.8.8.8: seq=4 ttl=255 time=27.975 ms
64 bytes from 8.8.8.8: seq=5 ttl=255 time=27.402 ms
^C
--- 8.8.8.8 ping statistics ---
7 packets transmitted, 6 packets received, 14% packet loss
round-trip min/avg/max = 26.690/27.596/28.318 ms
root@OpenWrt:~# ping google.com
PING google.com (192.178.158.100): 56 data bytes
64 bytes from 192.178.158.100: seq=0 ttl=255 time=28.663 ms
64 bytes from 192.178.158.100: seq=1 ttl=255 time=28.525 ms
64 bytes from 192.178.158.100: seq=2 ttl=255 time=28.480 ms
64 bytes from 192.178.158.100: seq=3 ttl=255 time=30.433 ms
64 bytes from 192.178.158.100: seq=4 ttl=255 time=29.078 ms
^C
--- google.com ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 28.480/29.035/30.433 ms
root@OpenWrt:~#

```

Figure 9: Successful Internet Connectivity Test from OpenWrt to External Networks

```

<?xml version="1.0" encoding="utf-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN" "http://www.w3.org/TR/xhtml11/
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <meta http-equiv="Cache-Control" content="no-cache, no-store, mu
    <meta http-equiv="refresh" content="0; URL=cgi-bin/luci/" />
    <style type="text/css">
      body { background: white; font-family: arial, helvetica,
        a { color: black; }

      @media (prefers-color-scheme: dark) {
        body { background: black; }
        a { color: white; }
      }
    </style>
  </head>
  <body>
    <a href="cgi-bin/luci/">LuCI - Lua Configuration Interface</a>
  </body>
</html>
- /www/index.html 1/20 5%

```

Figure 10: OpenWrt Web Interface HTML in /www/index.html

4.1.3 Configure Firewall Rules

```
C:\Users\admin>ssh root@192.168.56.2
The authenticity of host '192.168.56.2 (192.168.56.2)' can't be established.
ED25519 key fingerprint is SHA256:JU7JpdvC4sg7P+7Cn9xGgQoEdoxf5ZcHrKDepPIu7+U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.2' (ED25519) to the list of known hosts.
root@192.168.56.2's password:
Permission denied, please try again.
root@192.168.56.2's password:
Permission denied, please try again.
root@192.168.56.2's password:
C:\Users\admin>ssh root@192.168.56.2
root@192.168.56.2's password:
Permission denied, please try again.
root@192.168.56.2's password:
Permission denied, please try again.
root@192.168.56.2's password:
Connection closed by 192.168.56.2 port 22
```

Figure 11: Failed SSH Authentication Attempt on OpenWrt Through TCP Port 22

```
tcp      0      0 0.0.0.0:22          0.0.0.0:*           LISTEN
1691/dropbear
tcp      0      0 :::22              :::*                 LISTEN
1691/dropbear
root@OpenWrt:~# uci show dropbear
dropbear.@dropbear[0]=dropbear
dropbear.@dropbear[0].PasswordAuth='on'
dropbear.@dropbear[0].RootPasswordAuth='on'
dropbear.@dropbear[0].Port='22'
root@OpenWrt:~# uci add firewall rule
cfg0f92bd
root@OpenWrt:~# uci set firewall.@rule[-1].name='Allow-SSH-2222'
root@OpenWrt:~# uci set firewall.@rule[-1].src='lan'
root@OpenWrt:~# uci set firewall.@rule[-1].proto='tcp'
root@OpenWrt:~# uci set firewall.@rule[-1].dest_port='2222'
root@OpenWrt:~# uci set firewall.@rule[-1].target='ACCEPT'
root@OpenWrt:~# uci commit firewall
root@OpenWrt:~# /etc/init.d/firewall reload
root@OpenWrt:~# uci show firewall | grep -A6 'Allow-SSH-2222'
firewall.@rule[10].name='Allow-SSH-2222'
firewall.@rule[10].src='lan'
firewall.@rule[10].proto='tcp'
firewall.@rule[10].dest_port='2222'
firewall.@rule[10].target='ACCEPT'
root@OpenWrt:~#
```

Figure 12: OpenWrt Firewall Configuration Allowing SSH Traffic on TCP Port 2222

```

dropbear.@dropbear[0].Port='22'
root@OpenWrt:~# uci add firewall rule
cfg@f92bd
root@OpenWrt:~# uci set firewall.@rule[-1].name='Allow-SSH-2222'
root@OpenWrt:~# uci set firewall.@rule[-1].src='lan'
root@OpenWrt:~# uci set firewall.@rule[-1].proto='tcp'
root@OpenWrt:~# uci set firewall.@rule[-1].dest_port='2222'
root@OpenWrt:~# uci set firewall.@rule[-1].target='ACCEPT'
root@OpenWrt:~# uci commit firewall
root@OpenWrt:~# /etc/init.d/firewall reload
root@OpenWrt:~# uci show firewall | grep -A6 'Allow-SSH-2222'
firewall.@rule[10].name='Allow-SSH-2222'
firewall.@rule[10].src='lan'
firewall.@rule[10].proto='tcp'
firewall.@rule[10].dest_port='2222'
firewall.@rule[10].target='ACCEPT'
root@OpenWrt:~# uci set dropbear.@dropbear[0].port='2222'
root@OpenWrt:~# uci commit dropbear
root@OpenWrt:~# /etc/init.d/dropbear restart
root@OpenWrt:~# netstat -lntp | grep -E ':(22|2222)\b'
tcp        0      0 0.0.0.0:22                0.0.0.0:*               LISTEN
14951/dropbear
tcp        0      0 :::22                    :::*                     LISTEN
14951/dropbear
root@OpenWrt:~# _

```

```

C:\Users\admin>ssh -p 2222 root@192.168.56.2
ssh: connect to host 192.168.56.2 port 2222: Connection refused

```

Figure 13: SSH Port Modification to TCP 2222 and Connection Refusal During Port Verification



Example web page

Figure 14: Successful Access to the OpenWrt Web Server Through HTTP

```

Passwords don't match
passwd: password for root is unchanged
root@OpenWrt:~# passwd
Changing password for root
New password:
Bad password: too short
Retype password:
passwd: password for root changed by root
root@OpenWrt:~# cp /etc/config/firewall /etc/config/firewall.project-backup
root@OpenWrt:~# uci show firewall | grep -E "zone!network"
firewall.@zone[0]=zone
firewall.@zone[0].name='lan'
firewall.@zone[0].network='lan'
firewall.@zone[0].input='ACCEPT'
firewall.@zone[0].output='ACCEPT'
firewall.@zone[0].forward='ACCEPT'
firewall.@zone[1]=zone
firewall.@zone[1].name='wan'
firewall.@zone[1].network='wan' 'wan6'
firewall.@zone[1].input='REJECT'
firewall.@zone[1].output='ACCEPT'
firewall.@zone[1].forward='REJECT'
firewall.@zone[1].masq='1'
firewall.@zone[1].mtu_fix='1'
root@OpenWrt:~#

```

Figure 15: OpenWrt Root Password Hardening and Firewall Zone Configuration

```

uhttpd.main.rfc1918_filter='1'
uhttpd.main.max_requests='3'
uhttpd.main.max_connections='100'
uhttpd.main.cert='/etc/uhttpd.crt'
uhttpd.main.key='/etc/uhttpd.key'
uhttpd.main.cgi_prefix='/cgi-bin'
uhttpd.main.lua_prefix='/cgi-bin/luci=/usr/lib/lua/luci/cgi/uhttpd.lua'
uhttpd.main.script_timeout='60'
uhttpd.main.network_timeout='30'
uhttpd.main.http_keepalive='20'
uhttpd.main.tcp_keepalive='1'
uhttpd.main.ubus_prefix='/ubus'
uhttpd.defaults=cert
uhttpd.defaults.days='730'
uhttpd.defaults.key_type='ec'
uhttpd.defaults.bits='2048'
uhttpd.defaults.ec_curve='P-256'
uhttpd.defaults.country='ZZ'
uhttpd.defaults.state='Somewhere'
uhttpd.defaults.location='Unknown'
uhttpd.defaults.commonname='OpenWrt'
uhttpd.student=uhttpd
uhttpd.student.listen_http='80'
uhttpd.student.home='/srv/www'
root@OpenWrt:~# S

```

Figure 16: OpenWrt HTTP Service Configuration and TCP Port 80 Firewall Rule

```

uhttpd.defaults.ec_curve='P-256'
uhttpd.defaults.country='ZZ'
uhttpd.defaults.state='Somewhere'
uhttpd.defaults.location='Unknown'
uhttpd.defaults.commonname='OpenWrt'
uhttpd.student=uhttpd
uhttpd.student.listen_http='80'
uhttpd.student.home='/srv/www'
root@OpenWrt:~# uci add firewall rule
cfg@e92bd
root@OpenWrt:~# uci set firewall.@rule[-1].name='Block-http-port-80'
root@OpenWrt:~# uci set firewall.@rule[-1].src='lan'
root@OpenWrt:~# uci set firewall.@rule[-1].proto='tcp'
root@OpenWrt:~# uci set firewall.@rule[-1].dest_port='80'
root@OpenWrt:~# uci set firewall.@rule[-1].target='DROP'
root@OpenWrt:~# uci commit firewall
root@OpenWrt:~# /etc/init.d/firewall reload
root@OpenWrt:~# uci show firewall | grep -A6 'Block-HTTP'
root@OpenWrt:~# uci set firewall.@rule[-1].target='ACCEPT'
> ^C
root@OpenWrt:~# uci set firewall.@rule[-1].target='ACCEPT'
root@OpenWrt:~# uci commit firewall
root@OpenWrt:~# /etc/init.d/firewall reload
root@OpenWrt:~# uci show firewall | grep -A6 'Block-HTTP'
root@OpenWrt:~#

```

Figure 17: Successful ICMP Connectivity Test from Windows Host to OpenWrt

```

</html>

C:\Users\admin>ping 192.168.56.2

Pinging 192.168.56.2 with 32 bytes of data:
Reply from 192.168.56.2: bytes=32 time<1ms TTL=64
Reply from 192.168.56.2: bytes=32 time=1ms TTL=64
Reply from 192.168.56.2: bytes=32 time<1ms TTL=64
Reply from 192.168.56.2: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.56.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\admin>

```

Figure 18: OpenWrt Firewall and Network Security Configuration Verification

Firewall Filtering provides some extra security on the network, as it restricts access to key services. The block of port 80 prevents any unauthorised access of web services and then it is opened back to test the service availability in a controlled way. Administrative access through SSH port 22 is initially allowed and it is later switched to port 2222 thus lessening the chances of making SSH automatic attacks (Kuruppathukattil, 2025). Blocking ICMP helps prevent network reconnaissance, re-enabling it when it is needed for connectivity testing. The management interface port 81 is limited and access to it may not be provided to any unauthorised administrative access. Together, these rules mandate controlled connectivity, eases the attack surface, protects administrative services and enhances network security.

4.1.4 Network Diagram and Address Allocation Production

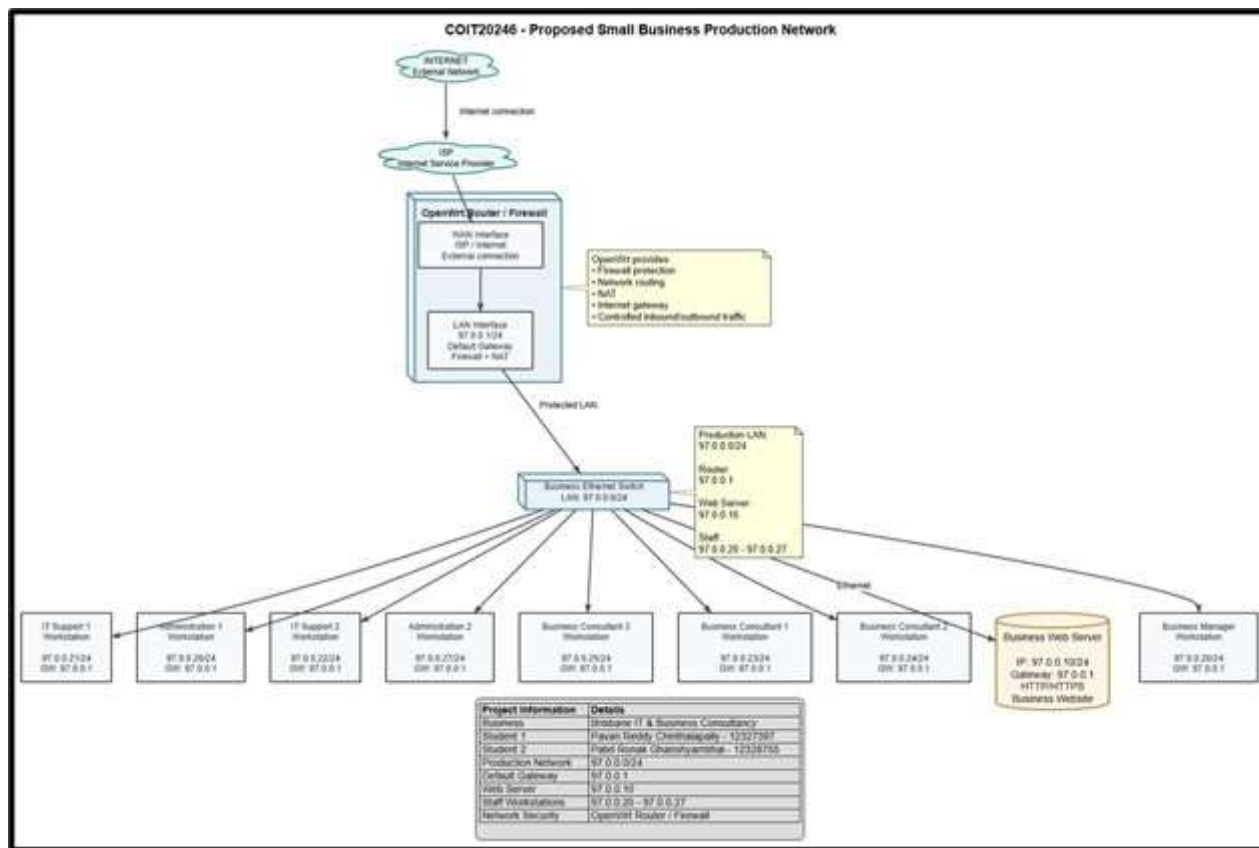


Figure 19: Proposed Small Business Production Network Topology and IP Address Allocation

The suggested production network includes a OpenWrt router/firewall for Internet connection and several routing and NAT services, thus offering traffic protection and routing to the Internet. The protected 77.0.0.0/24 LAN is used to physically connect the business Ethernet switch and the dedicated web server to 8 staff workstations. The web server uses 77.0.0.10, while staff devices use 77.0.0.20--77.0.0.27 (Gentile et al. 2024). The default gateway for the router is 77.0.0.1. The structured design facilitates safe connectivity, managed network access, business website functionality and resource management.

4.1.5 IP Addressing Requirements

Device / Role	IP Address	Subnet Mask	Default Gateway	Purpose
OpenWrt Router/Firewall	77.0.0.1	255.255.255.0 (/24)	---	Network gateway, routing and firewall
Business Web Server	77.0.0.10	255.255.255.0 (/24)	77.0.0.1	Business website

Device / Role	IP Address	Subnet Mask	Default Gateway	Purpose
Business Manager	97.0.0.20	255.255.255.0 (/24)	97.0.0.1	Staff workstation
IT Support 1	97.0.0.21	255.255.255.0 (/24)	97.0.0.1	Staff workstation
IT Support 2	97.0.0.22	255.255.255.0 (/24)	97.0.0.1	Staff workstation
Business Consultant 1	97.0.0.23	255.255.255.0 (/24)	97.0.0.1	Staff workstation
Business Consultant 2	97.0.0.24	255.255.255.0 (/24)	97.0.0.1	Staff workstation
Business Consultant 3	97.0.0.25	255.255.255.0 (/24)	97.0.0.1	Staff workstation
Administration 1	97.0.0.26	255.255.255.0 (/24)	97.0.0.1	Staff workstation
Administration 2	97.0.0.27	255.255.255.0 (/24)	97.0.0.1	Staff workstation

4.2 Security Hardening and Traffic Analysis

4.2.1 Harden the OpenWRT System

```
C:\Users\admin>ssh root@192.168.56.2  
root@192.168.56.2's password:  
  
BusyBox v1.35.0 (2023-01-03 00:24:21 UTC) built-in shell (ash)  
  
 _ _ | W I R E L E S S   F R E E D O M  
-----  
OpenWrt 22.03.3, r20028-43d71ad93e  
-----  
root@OpenWrt:~#
```

Figure 20: Successful SSH Login to the OpenWrt Virtual Machine

```

tcp 0 0 0.0.0.0:22 0.0.0.0:22 LISTENING
14951/dropbear
root@OpenWrt:~# passwd
Changing password for root
New password:
Bad password: too short
Retype password:
passwd: password for root changed by root
root@OpenWrt:~#

```

Figure 21: Successful Root Password Change on OpenWrt

```

root@OpenWrt:~# cat /etc/shadow
root:$1$USpcxGb5$RX1bJlMrQiaWjRGp6jPha1:20728:0:99999:7:::
daemon:*:0:0:99999:7:::
ftp:*:0:0:99999:7:::
network:*:0:0:99999:7:::
nobody:*:0:0:99999:7:::
ntp:x:0:0:99999:7:::
dnsmasq:x:0:0:99999:7:::
logd:x:0:0:99999:7:::
ubus:x:0:0:99999:7:::
root@OpenWrt:~#

```

Figure 22: OpenWrt Password Hashes Stored in /etc/shadow

```

C:\Users\admin>ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\admin/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Passphrases do not match. Try again.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\admin/.ssh/id_ed25519
Your public key has been saved in C:\Users\admin/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:5VEISV5CTpdVRRRCFucLe4R+d0KrQlXpSuPy14tJlMRE tech assignment 02@Administrator
The key's randomart image is:
+--[ED25519 256]--+
  =O..+OE=O+
  + OO..O
  + +..+
  . = OOBO.
  S +..+..O
  . *..O+..
  O B O..
  . + O .
  O..
+----[SHA256]-----+

C:\Users\admin>

```

```

root@OpenWrt:~# cat /etc/shadow
root:$1$USpcxGb5$RX1bJlMrQiaWjRGp6jPha1:20728:0:99999:7:::
daemon:*:0:0:99999:7:::
ftp:*:0:0:99999:7:::
network:*:0:0:99999:7:::
nobody:*:0:0:99999:7:::
ntp:x:0:0:99999:7:::
dnsmasq:x:0:0:99999:7:::
logd:x:0:0:99999:7:::
ubus:x:0:0:99999:7:::
root@OpenWrt:~#

```

Figure 23: Ed25519 SSH Key Pair Generation on the Windows Host

Key-based authentication is more secure because it uses a cryptographic key pair instead of relying solely on passwords. The private key remains securely on the authorised device, while the public key is stored on OpenWrt. This reduces exposure to password guessing, brute-force attacks, credential theft, and password reuse.

```

/network
lrwxrwxrwx    1 root    root    16 Jan  3  2023 S35odhcpd -> ../init.d/
odhcpd
lrwxrwxrwx    1 root    root    14 Jan  3  2023 S50cron -> ../init.d/cr
on
lrwxrwxrwx    1 root    root    16 Oct  2  06:37 S50uhttpd -> ../init.d/
uhttpd
lrwxrwxrwx    1 root    root    18 Jan  3  2023 S80ucitrack -> ../init.
d/ucitrack
lrwxrwxrwx    1 root    root    21 Jan  3  2023 S94gpio_switch -> ../in
it.d/gpio_switch
lrwxrwxrwx    1 root    root    14 Jan  3  2023 S95done -> ../init.d/do
ne
lrwxrwxrwx    1 root    root    13 Jan  3  2023 S96led -> ../init.d/led
lrwxrwxrwx    1 root    root    17 Jan  3  2023 S98sysntpd -> ../init.d
/sysntpd
lrwxrwxrwx    1 root    root    22 Jan  3  2023 S99urandom_seed -> ../i
nit.d/urandom_seed
root@OpenWrt:/# ls /etc/init.d
boot      firewall  odhcpd    system    urngd
cron      gpio_switch  rpcd      ucitrack
dnsmasq   led       sysctl    uhttpd
done      log       sysfixtime  umount
dropbear  network   sysntpd    urandom_seed
root@OpenWrt:/#

```

Figure 24: OpenWrt Enabled Services and Startup Configuration

Disabling unnecessary services reduces the system's attack surface by removing functions that are not required. Fewer active services mean fewer open ports, processes, and potential vulnerabilities for attackers to exploit. This limits possible entry points, reduces security risks, and makes the OpenWrt system easier to monitor and maintain securely.

4.2.2 Capture and Analyse Network Traffic

```

root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/# tcpdump -i br-mng -s 0 -w /tmp/http_capture.pcap host 192.168.56
.1 and tcp port 80
[ 6929.479623] device br-mng entered promiscuous mode
tcpdump: listening on br-mng, link-type EN10MB (Ethernet), capture size 262144 b
ytes
^C0 packets captured
0 packets received by filter
0 packets dropped by kernel
[ 7069.406559] device br-mng left promiscuous mode
root@OpenWrt:/#

```

Figure 25: HTTP Traffic Capture Using tcpdump on the OpenWrt br-mng Interface

```

root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/# tcpdump -i br-mng -s 0 -w /tmp/http_capture.pcap host 192.168.56
.1 and tcp port 80
[ 6929.479623] device br-mng entered promiscuous mode
tcpdump: listening on br-mng, link-type EN10MB (Ethernet), capture size 262144 b
ytes
^C0 packets captured
0 packets received by filter
0 packets dropped by kernel
[ 7069.406559] device br-mng left promiscuous mode
root@OpenWrt:/# S_

```

Figure 26: HTTP Traffic Capture Using tcpdump with Host and TCP Port Filtering

9	22.000032	192.168.56.1	192.168.56.255	UDP	82 64846 → 194/ Len=40
10	22.001480	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Leave group 224.168.100.1
11	22.488650	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Leave group 224.168.100.1
12	23.002769	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.168.100.1 for any sources
13	23.488548	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.168.100.1 for any sources
14	28.312307	192.168.56.1	239.255.255.250	SSDP	212 M-SEARCH * HTTP/1.1
15	29.312964	192.168.56.1	239.255.255.250	SSDP	212 M-SEARCH * HTTP/1.1
16	30.313835	192.168.56.1	239.255.255.250	SSDP	212 M-SEARCH * HTTP/1.1
17	31.314423	192.168.56.1	239.255.255.250	SSDP	212 M-SEARCH * HTTP/1.1

Figure 27: Captured Network Traffic Displayed in Wireshark (HTTP)

The attacker who taps information from an HTTP stream is able to obtain the server and client IP addresses, the request and response URL, HTTP requests and responses, and the content of the target web pages. Personal information, including names, student IDs or details that could be seen from the webpage, might be given away due to lack of encryption in the HTTP. Any traffic monitoring, and information disclosure, would be possible.

```

ip: can't find device br-mng
root@OpenWrt:/# tcpdump -i br-mng -s 0 -w /tmp/ssh_capture.pcap host 192.168.56.
1 and tcp port 22
[ 7647.756077] device br-mng entered promiscuous mode
tcpdump: listening on br-mng, link-type EN10MB (Ethernet), capture size 262144 b
ytes
^C0 packets captured
0 packets received by filter
0 packets dropped by kernel
[ 7691.350391] device br-mng left promiscuous mode
root@OpenWrt:/#

```

Figure 28: SSH Traffic Capture Attempt Using tcpdump on OpenWrt

56	126.282288	0a:00:27:00:00:04	LLDP_Multicast	LLDP	65	LA/ADMINISTRATOR MA/0a:00:27:00:00:04 3601
57	131.606133	192.168.56.1	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.168.100.1
58	132.000284	192.168.56.1	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.168.100.1
59	132.606697	192.168.56.1	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.168.100.1 for any sources
60	133.000242	192.168.56.1	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.168.100.1 for any sources
61	142.606635	192.168.56.1	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.168.100.1
62	143.000234	192.168.56.1	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.168.100.1
63	143.607263	192.168.56.1	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.168.100.1 for any sources
64	144.000388	192.168.56.1	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.168.100.1 for any sources

Figure 29: SSH Network Traffic Packets Displayed in Wireshark During Traffic Analysis

In the HTTP capture it is possible to read webpage content and requests. Encrypted protocols like SSH, on the other hand, keep contents of the sessions secure from being viewed directly. Encryption ensures the confidentiality of the data being sent, minimising the chances of attackers being able to obtain commands, credentials or any other sensitive information through network interception.

Hardening Summary

Hardening Step	Security Risk Addressed
1. Change Default Root Password	Reduces the risk of unauthorised administrative access through default or easily guessed credentials. A strong, unique password makes credential-based attacks more difficult.
2. Examine Password Storage in /etc/shadow	Addresses the risk of password disclosure. Storing passwords as hashes rather than plaintext prevents the original passwords from being directly exposed if the password database is accessed.
3. Set Up SSH Key-Based Authentication	Reduces the risk of password guessing and brute-force attacks. SSH keys provide stronger authentication than password-only access, particularly when a passphrase protects the private key.
4. Disable Unnecessary Services	Reduces the attack surface of OpenWrt. Unnecessary services can provide additional entry points that an attacker could exploit.

4.3 Risk Assessment and Security Controls

4.3.1 Conduct a Cyber Security Risk Assessment

ID	Candidate threat / vulnerability	Scenario-specific risk	Primary asset	Asset type	Data asset	Existing control/evidence	Likelihood (1-5)	Impact (1-5)	Inherent score	Treatment / control
R01	Unauthorised access / credential compromise	Weak or compromised administrator credentials could allow privileged access to the network.	A05 Employee credentials	Information / credential	Y	Root password changed; SSH key authentication	4	5	20	Use unique strong credentials, Ed25519 keys and restrict management access.
R02	Network interception / eavesdropping	Unencrypted HTTP exposes webpage content and network metadata to an interceptor.	A06 Business service/content information	Information / data	Y	HTTP capture demonstrated plaintext content; SSH is encrypted.	4	4	16	Use HTTPS/TLS for the production website and avoid sensitive data over HTTP.
R03	Malware / endpoint compromise	A compromised Windows workstation could expose business records or provide a route into the internal network.	A02 Windows staff workstations	Hardware / endpoint	N	Firewall and OpenWrt gateway; no endpoint security evidence recorded	3	5	15	Endpoint protection, patching, least privilege and controlled application installation.
R04	Denial of service / service disruption	Flooding or excessive requests against the public web service could reduce website availability.	A03 Business web server/httpsd	Software / service	N	Firewall configured for traffic control	3	4	12	Rate limiting, monitoring, backup web hosting and restrictive firewall policy.
R05	Unauthorised modification	An attacker gaining administrative access could alter firewall or web-server configuration.	A09 OpenWrt firewall configuration	Software / configuration	N	SSH moved to 2222; key authentication; root password changed	3	5	15	Restrict administration, back up configuration, monitor changes and use least privilege.
R06	Information disclosure	Business/client documents or contact information could be exposed from compromised endpoints or storage.	A07 Business documents and client records	Information / data	Y	Network firewall; SSH encryption; hardening	3	5	15	Access control, encryption at rest/in transit, backups and staff awareness.
R07	Social engineering / phishing	Staff could disclose credentials or open malicious content, leading to account or endpoint compromise.	A10 Staff and IT support personnel	People	N	No phishing-specific control demonstrated	3	4	12	Security awareness training, phishing simulations and MFA where supported.
R08	Physical theft / loss	Theft of a workstation or administrator device could expose business data or SSH private keys.	A08 SSH private keys	Information / credential	Y	Private key generated with passphrase	2	5	10	Device encryption, secure storage, key backup/revocation and physical access controls.
R09	Unnecessary service exposure	An unused network service could create an additional attack surface.	A01 OpenWrt router/firewall	Hardware / network infrastructure	N	Unnecessary service disabled as part of hardening	3	4	12	Disable unused services and periodically review listening ports.
R10	ICMP/network reconnaissance	Exposed ping/ICMP can assist network discovery, although it can also support diagnostics.	A01 OpenWrt router/firewall	Hardware / network infrastructure	N	ICMP block/allow rule demonstrated	3	2	6	Permit ICMP only where operationally required and monitor unusual traffic.
R11	SSH service exposure	Leaving administration exposed on a standard port can attract automated scanning and brute-force attempts.	A01 OpenWrt router/firewall	Hardware / network infrastructure	N	SSH moved from 22 to 2222; port 22 blocked; key authentication	3	4	12	Restrict SSH source addresses, use keys, monitor logs and keep software patched.
R12	Web management interface exposure	Management web access could expose administrative functions if reachable from untrusted networks.	A01 OpenWrt router/firewall	Hardware / network infrastructure	N	Assignment requires management-interface restriction	3	5	15	Permit management only from trusted management network and verify port 81 access restrictions.

Figure 30: Cybersecurity Risk Assessment Matrix and Recommended Controls

A mini business network developed in Sections 4.1 and 4.2 is subject to a mini cyber security risk assessment, carried out with the TVAMatrix template. The assessment included 12 types of information security threats such as: unauthorised access, network interception, malware, denial of service, unauthorised modification, information disclosure, phishing, physical theft, unnecessary service exposure, ICMP reconnaissance, SSH exposure, management-interface exposure. The assets covered hardware, software/services, information/data, credentials and the people. These were business/client records, business/Service information, employee credentials and SSH private keys. The controls that already existed were firewall rules, SSH key authentication, hardening of the root password, encryption and turning off unnecessary services.

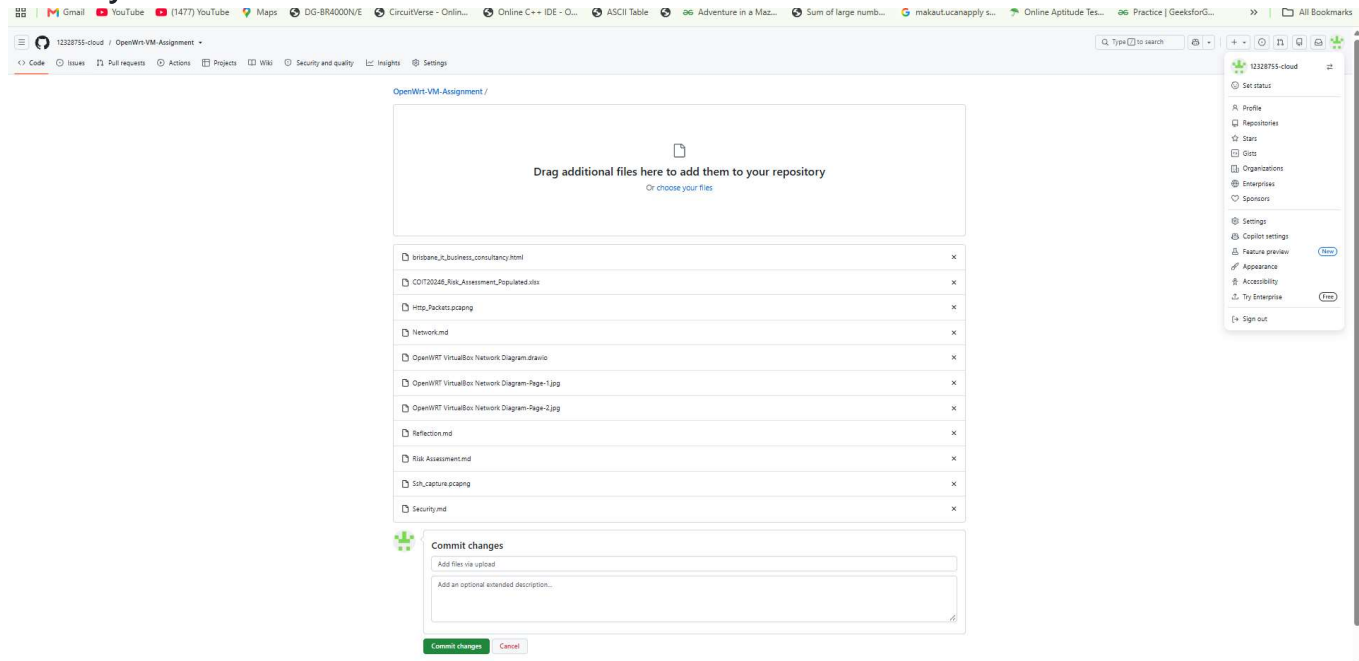
4.3.2 Recommend Security Controls

Employee credentials (A05) is the highest-risk data asset, with an inherent risk score of 20. There are three suggested security controls:

1. Multi-Factor Authentication (MFA): This decreases the likelihood of a credential being compromised. Must be activated for administrators and employees. This could result in longer logon, and use of an additional authentication mechanism will be necessary for users.
2. Strong Authentication and SSH Key Management: Use of SSH Key Management is highly encouraged and SSH keys are going to replace password access, where the private key is protected with a passphrase (Makris, 2024). Moving SSH from its own port 22 and hardening its access with port 2222 means less exposure from a brute-force approach. The key management may cause recovery and admin overhead.

3. Access Control/Firewall Restrictions: Administrative services should be networked within a management network and unneeded ports should be closed. A foundation of exit and entry firewall rules for SSH, HTTP and ICMP and management.

4.4 Project Reflection



Project Reflection -- Task Allocation and Contribution Table

Student	Student ID	Actual Tasks Completed	GitHub Contribution
Pavan Reddy Chinthalapally	12327397	OpenWrt VM configuration; network interface and connectivity testing; firewall configuration and testing; SSH hardening; password and key authentication; tcpdump /Wireshark traffic analysis; risk assessment support; report documentation.	Commits related to OpenWrt configuration, security testing, traffic analysis and documentation.
Patel Ronak Ghanshyambhai	12328755	Network planning and diagram development; website configuration/testing; firewall and connectivity support; risk assessment and security controls; report preparation;	Commits related to network design, website, risk assessment, documentation and project organisation.

Student	Student ID	Actual Tasks Completed	GitHub Contribution
		GitHub organisation and supporting documentation.	

I configured the OpenWrt virtual machine, tested network connectivity, wrote firewall rules, system hardening, captured network traffic with tcpdump and wireshark and assisted in the risk assessment. My teammate was responsible for other work configuration, work documentation, testing and work on GitHub. Our individual effort was demonstrated in our commits to the Git database; however, we did not always have the top number of commits for the highest effort since some were done in larger, more fundamental commits and others in smaller, more granular ones. Things were done during the project weeks and we committed. The group was very good on communication and in sharing the testing, some delays were depends on configuration errors. For future projects, the use of regular task allocation, progress meetings, and more explicit communication in the Git commit messages needs to be used to enhance coordination and accountability.

References

Gentile, A.F., Macrì, D., Greco, E. and Fazio, P., 2024. IoT IP overlay network security performance analysis with open source infrastructure deployment. *Journal of Cybersecurity and Privacy*, 4(3), pp.629-649.

Kuruppathukattil, V., 2025, August. Optimized WiFi Network Deployment with OpenWRT and FreeRadius for Secure and Scalable Connectivity. In *2025 IEEE 2nd International Conference on Information Technology, Electronics and Intelligent Communication Systems (ICITEICS)* (pp. 1-6). IEEE.

Makris, N., 2024. Cyber range development: Configuration of the cyber range environment network and monitoring tools (Master's thesis, Πανεπιστήμιο Πειραιώς).